

/toolchain

Language toolchain report: `/toolchain list [lang] [--json] | /toolchain help`

Overview

`/toolchain` reports the presence of language toolchains detected on this machine: which runtimes and compilers are installed, their versions, and their readiness status. It is strictly read-only - it never installs, upgrades, or configures anything; it only probes. For example, `/toolchain list rust` probes cargo and rustc and reports what was found.

Syntax

```
/toolchain
/toolchain help
/toolchain list
/toolchain list <language>
/toolchain list <language> --json
/toolchain list --json
```

Notes:

- The bare form is equivalent to `/toolchain help`.
- `--json` is position-independent: it may appear before or after the language filter.
- The language filter is case-insensitive.
- Probing is blocking; while it runs the status line shows `[wait] toolchain status`.

Options / Subcommands

Form	Description
<code>/toolchain</code> (or <code>/toolchain help</code>)	Print the help page.
<code>/toolchain list</code>	Probe and report every known language toolchain.
<code>/toolchain list <language></code>	Probe and report only the given language (case-insensitive id).
<code>/toolchain list <language> --json</code>	Same report as a bare JSON document.
unknown language id	Unknown language id{filter}. Valid ids: ...
unknown subcommand	Unknown subcommand{sub}. Usage: <code>/toolchain [list [language] [--json] help]</code> . Run <code>/toolchain help</code> for details.

Examples

Probe everything and show the plain table:

```
/toolchain list
```

Probe only the Rust toolchain (cargo + rustc):

```
/toolchain list rust
```

Machine-readable report for scripts or other tools:

```
/toolchain list rust --json
```

Flag position does not matter:

```
/toolchain list --json rust
```

Unknown subcommand prints the usage correction:

```
/toolchain install rust
```

Output

Table form

A fixed-width table (constant 93-column width) with 10/10/10/50 character columns:

Column	Width	Overflow
Language	10	clipped
Runtime	10	clipped
Status	10	word-wrapped
Version	50	word-wrapped

Wrapped rows continue on grid lines, so multi-word statuses and long versions remain readable inside the fixed grid.

JSON form

`--json` renders a bare JSON document (no table) with the same probe results, one entry per toolchain, suitable for piping into other tools.

Errors and status

- While probing: `status [wait] toolchain status`.
- Unknown language filter: `Unknown language id{filter}`. Valid ids: ...
- Unknown subcommand: `Unknown subcommand{sub}`. Usage: `/toolchain [list [language] [--json] | help]`. Run `/toolchain help` for details.

The command never installs or upgrades toolchains; missing toolchains simply report as not found.

Related

- Toolchain manual: `docs/howtos/toolchain.md`.
- README v1.0.95 notes the fixed-width `/toolchain list` table columns.
- `/new` scaffolding detects the runtime stack it needs; `/doctor` and `/update` cover other environment checks.